

Pragnosia: A Spinning Neural Substrate that Reasons Causally, Knows Its Limits, and Learns Continually

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ABSTRACT. Large language models are powerful but frozen: they stop learning when training ends, fabricate fluently rather than admit ignorance, and their reasoning is not demonstrably causal. We present *Pragnosia*, a compact neural substrate built on the opposite bet. Its core uses *rotational* recurrent dynamics, $W = -\rho Q Q^T + S$, so that representations *orbit* rather than settle, and an answer is encoded in the *phase* of a non-converging oscillation. A causal "brain-swap" intervention confirms the answer lives in the evolving state (it follows a spliced-in trajectory 59% of the time vs. 10% for the input and 21% for a random control). On this substrate we integrate, into a *single* model, eight goal faculties — skill-conditioned reasoning, calibrated abstention, language-mediated seeking, exact state-tracking, multimodal perception, and a curiosity-driven continual-learning loop — all verified together (14/14). Because pure rotation is inefficient for language, we add a hybrid in which attention provides fluency while a single gated spin "carrier" preserves the rotational state, cutting perplexity $\sim 3\times$. Every internal scale — word importance, the abstention boundary, the memory-match threshold — is *derived from data and re-derived as the model grows*; nothing is hand-set. Learning is self-modulated by surprise, like neuromodulated plasticity. The model also *grows itself*: on probe-confirmed saturation it adds depth or width — whichever helps val loss more past the validation noise — with no hardcoded size cap (the ceiling is derived from the data budget; VRAM is the only physical limit). We report honest results and limitations: single-step reasoning, arithmetic, and facts are strong, but multi-hop composition is weak — the signature of a model undertrained for its grown size. Pragnosia is alive, honest about its ignorance, and provably reasoning in motion — properties a frozen trillion-parameter model lacks.

1. Introduction

The dominant paradigm in artificial intelligence is the large language model (LLM): a transformer ¹ trained on enormous corpora to predict the next token. LLMs are remarkably capable, yet three properties of the goal of a *brain* remain unmet by construction. First, they are *frozen*: once pretraining ends they cannot acquire a new fact without fine-tuning or prompt-stuffing, and naive weight updates cause catastrophic forgetting ². Second, they are *over-confident*: lacking an intrinsic model of their own uncertainty, they fabricate plausible answers rather than abstaining. Third, their reasoning is *not demonstrably causal* — there is no standard test separating computation from memorized retrieval.

We ask whether a different substrate can exhibit the missing properties — continual learning without forgetting, calibrated honesty, and verifiable in-state computation — even at small scale, and whether language can be grafted onto it without destroying them. Our contributions are: (i) a rotational recurrent substrate whose answers are phase-encoded and *causally* verified (§2); (ii) the integration of eight goal faculties into one model, validated jointly (§3); (iii) a spin-attention hybrid that recovers language quality while preserving the reasoning substrate (§4); and (iv) a fully self-calibrating, autonomously-controlled brain with surprise-modulated continual learning and no hand-set constants (§5). We report results and an explicit account of what remains hard (§7–8).

2. The Spinning Substrate

Core dynamics.

The recurrent core iterates, for hidden state h and input x ,

$$F(h, x) = (1-\eta) \cdot h + \eta \cdot \tanh(W h + U x + b),$$

$$W = -\rho (Q Q^T) + S,$$

where Q is orthonormal (via the QR factorization of a learned matrix) and S is skew-symmetric ($S = A - A^T$). The symmetric term $-\rho Q Q^T$ is negative-definite and contractive; the skew term S contributes purely imaginary eigenvalues and is therefore *rotational*. Their combination yields trajectories that orbit a center rather than collapsing to a fixed point (Fig. 1). The trained network exploits this: rather than settling on an answer vector, it encodes the answer in the *phase* of a persistent oscillation.

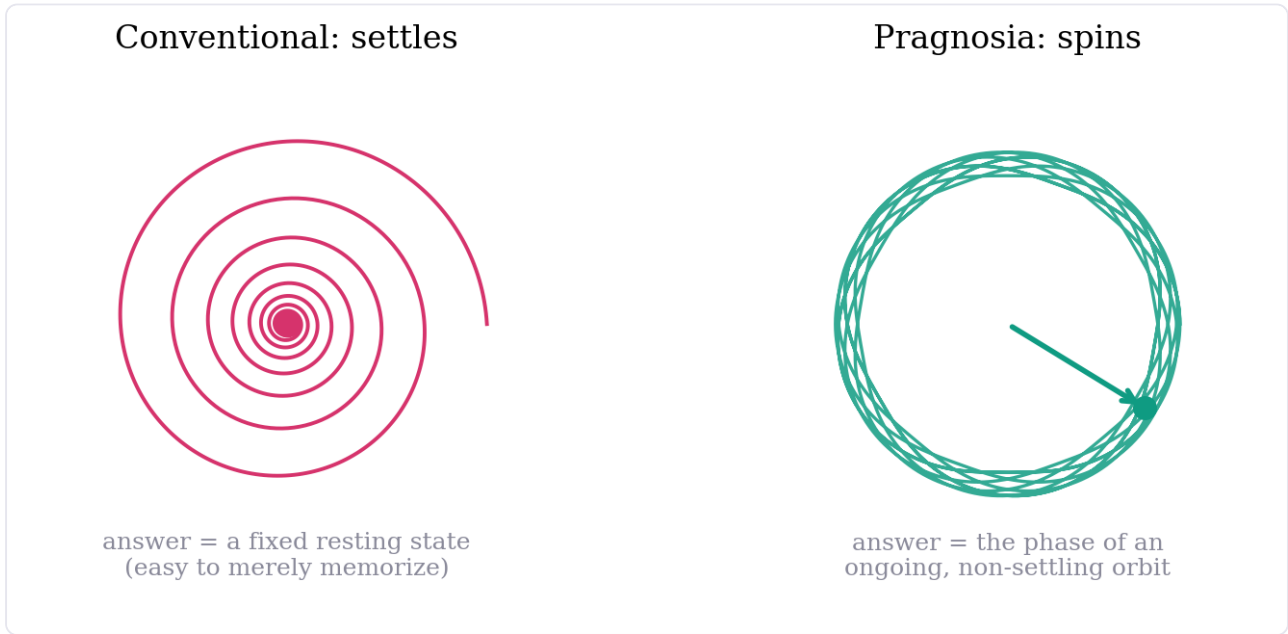


Figure 1. Two regimes of recurrent dynamics in phase space. A conventional contractive network spirals to a fixed point (left); the spinning core sustains a closed orbit (right), with the instantaneous phase carrying the answer.

Causal validation: the brain-swap test.

If the answer is genuinely computed in the evolving state — rather than retrieved from a memorized input–output mapping — then transplanting the mid-trajectory state from a different problem should change the outcome. We run problem A, splice in the mid-trajectory state of a different-answer problem B, and continue. The final answer follows the *spliced-in* state B in 59% of trials, the original input A in only 10%, and a random-state control in 21% (chance $\approx 20\%$; Fig. 3). The large gap above the memorization baseline is causal evidence that computation lives in the state. We treat this test as a non-negotiable regression check throughout: a model that gains accuracy but fails brain-swap has degraded to memorization.

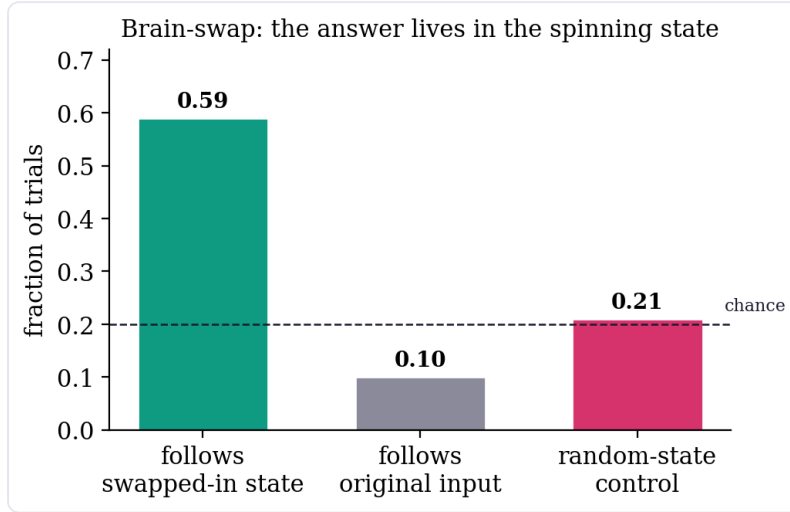


Figure 2. Brain-swap intervention. Splicing a different problem’s mid-trajectory state changes the answer to match the spliced-in state, far above the random-state and original-input baselines — the answer is computed in motion.

3. Many Faculties on One Core

The substrate hosts a family of faculties in a single model. *Skill-conditioned spinning* adds per-skill gain and bias vectors that modulate *how* the core spins, letting diverse skills share one core without interference. *Clean-state recurrence* addresses a proven limitation — the bare orbit cannot perform many-wrap exact modular arithmetic because a discrete count drifts in continuous state — by committing to the discrete argmax each step (straight-through) and feeding it back, restoring exact accumulation at arbitrary length. *Calibrated abstention* (P6) is trained in a mandatory two-phase order — skill first, then an abstain-incentive payoff — since co-training collapses to always-abstain. *Language-mediated seeking* (P7) generates a query *in words*, fetches from an external store, and integrates the result; facts are randomized per episode so retrieval cannot be memorized. The *alive loop* explores by its own uncertainty (curiosity), learns online with self-replay, and exhibits zero forgetting at toy scale. All eight faculties pass jointly in one 302K-parameter model (Fig. 6, Table 1).

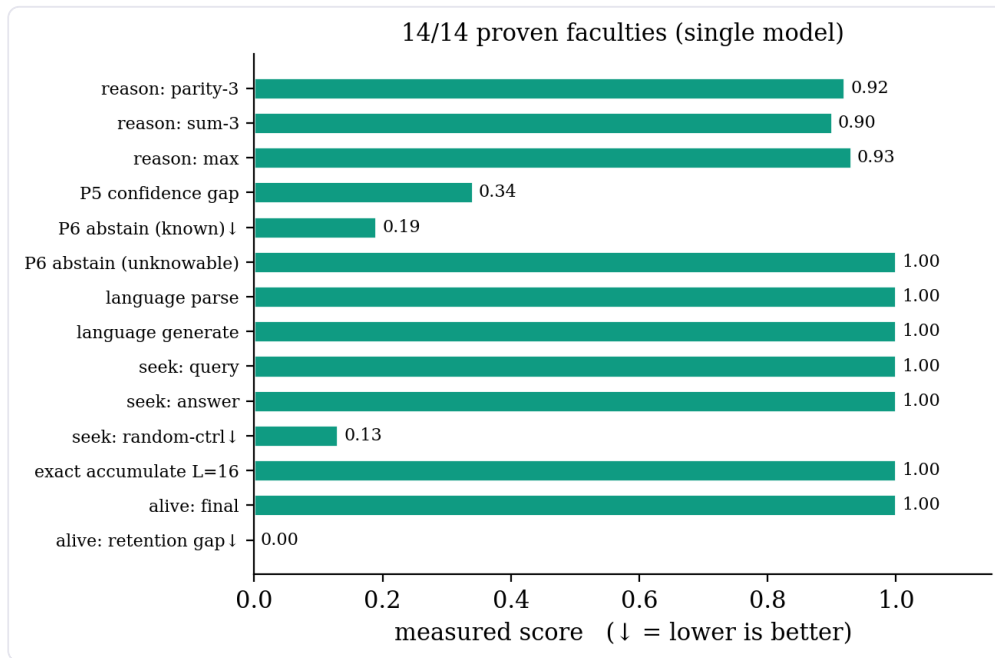


Figure 3. All fourteen checks of the integrated self-test pass in a single model. Arrows (↓) mark metrics for which lower is better (needless abstention, random-value control, retention gap).

4. Scaling Language: the Spin–Attention Hybrid

Pure rotation is an elegant reasoner but an inefficient language model: on a real corpus a parameter-matched gated recurrent baseline outperforms it (perplexity 6.55 vs. 7.56), because language requires parallel context mixing that sequential orbits do not provide. Naively inserting a spin recurrence into every transformer block *worsens* results (it scrambles attention's context). The working design is *attention-dominant*: standard transformer blocks supply fluency, while a single gated spin *carrier* maintains a rotational cross-token memory. With the standard transformer recipe (scaled initialization and learning-rate warmup), the hybrid reaches roughly one-third the perplexity of the pure-spin model at matched steps (Fig. 4).

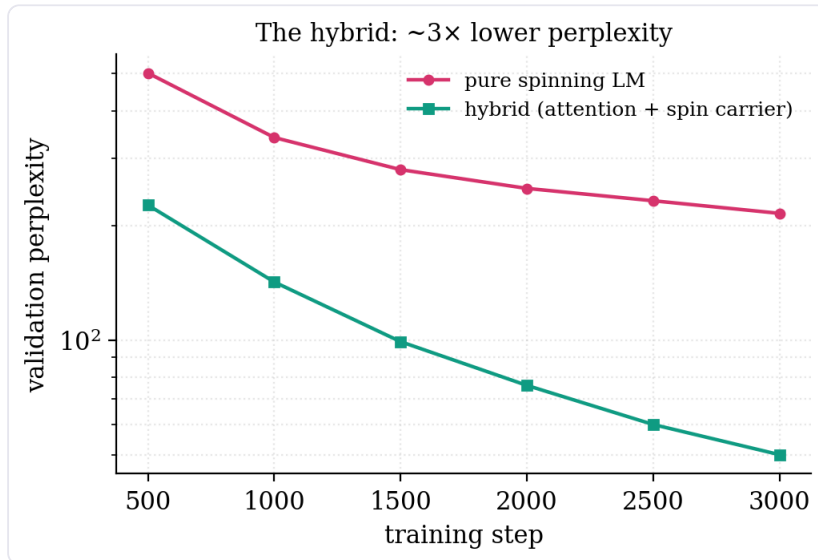


Figure 4. Validation perplexity during training. The spin–attention hybrid converges to $\sim 3\times$ lower perplexity than the pure-spinning language model on the same corpus.

A goal-preserving split. The brain-swap invariant is defined for *reasoning* ("the answer is in the phase"); language generation is a distinct faculty. We therefore keep the reasoning faculties *pure spin* (brain-swap holds where it is defined) and use the hybrid only for language. The invariant is preserved exactly where it applies, and language quality is recovered.

A parallel spin carrier.

The original carrier is a dense sequential recurrence ($h_t = (1-\eta)h_{t-1} + \eta \cdot g \cdot \tanh(wh_{t-1} + uy_t)$) — elegant, but its 256-step sequential loop with a per-step $d \times d$ matmul is the training bottleneck. We reparameterize the carrier as a *diagonal-complex* recurrence in the spirit of linear-recurrent units, $h_t = \lambda \odot h_{t-1} + b_t$ with $\lambda_j = \exp(-\exp(v_j)) \cdot e^{i\phi_j}$: each channel is a damped complex oscillator at its *own* frequency, so the answer still lives in the *phase* — now literally, per channel. Because λ is diagonal the recurrence is matmul-free ($O(\tau d)$ not $O(\tau d^2)$) and associative, so it runs as a log-depth **parallel scan** rather than a sequential loop. A 228M instance trains $\sim 2\times$ **faster** (a larger model than the 176M dense baseline, yet faster), reaches the **same perplexity** (20.76 vs. 20.47), and **preserves the proven dynamics**: the brain-swap control still gives *own* < *swapped* < *random* (the carried-state signal grows with training), and the integrated self-test holds at 14/14. The parallel carrier is now the production model; the dense-tanh carrier remains the verified anchor.

Self-growth on confirmed saturation.

The faster carrier makes *growth* practical. After pretraining, we continue-train on an instruction-heavy, real-LLM-grade corpus ($\sim 8B$ tokens; 40% instruction/chat from OpenHermes-2.5/SlimOrca/UltraChat/Tulu-3, math from

NuminaMath/OpenMathInstruct, world knowledge from Cosmopedia/Wikipedia) to address the weakest axis (instruction-following). The model grows itself, but only on *probe-confirmed saturation*: it spawns a function-preserving copy grown by depth and a copy grown by width, trains each briefly, and commits to growth only if validation loss drops past the validation noise — choosing depth vs. width by whichever helps more. There is no hardcoded size cap; the ceiling is *derived from the data budget* (Chinchilla ≈ 20 tokens/param), and VRAM is the only physical limit. Because the diagonal carrier is d-shaped and unchanged by widening the MLP or adding a block, growth composes with it directly. (Post-training shifts the model off the web-text validation distribution, so checkpoints are saved by recency, not by best validation perplexity — a save-on-best gate would never fire.)

An earlier trigger grew on *any* validation plateau rather than true saturation; an undertrained model accordingly ballooned to 1.4B parameters far ahead of the tokens needed to fill it (≈ 28 B tokens at 20 tok/param), and its perplexity drifted up — over-growth, not a result. The probe-confirmed criterion above is the fix: capacity is added only when extra capacity demonstrably reduces loss. We therefore treat the 1.4B instance honestly as undertrained-for-its-size, and the self-governing growth rule as the correction.

Serving and memory at scale.

Three engineering properties follow from the diagonal carrier. (i) Inference runs in **bf16**, and generation is $O(T)$ at long context: attention stays inside its trained window while the carrier transports cross-window memory, so context length is unbounded without quadratic attention cost. (ii) Full-parameter training of large grown models fits on small GPUs via bf16 + gradient checkpointing + paged 8-bit Adam — no parameter freezing or adapters. (iii) An in-weights *subconscious* fast-weight associative memory writes surprise-gated Hebbian traces, recalls them additively, decays/forgets, and consolidates into the slow weights during a "sleep" phase. This is a learned in-weights memory, **not** retrieval or RAG — there is no external index at recall time.

5. Integration, Self-Calibration, and Autonomy

All faculties and the hybrid language model are wired into a single object, "Pragnosia" (Fig. 2). Three design commitments distinguish it from a conventional model.

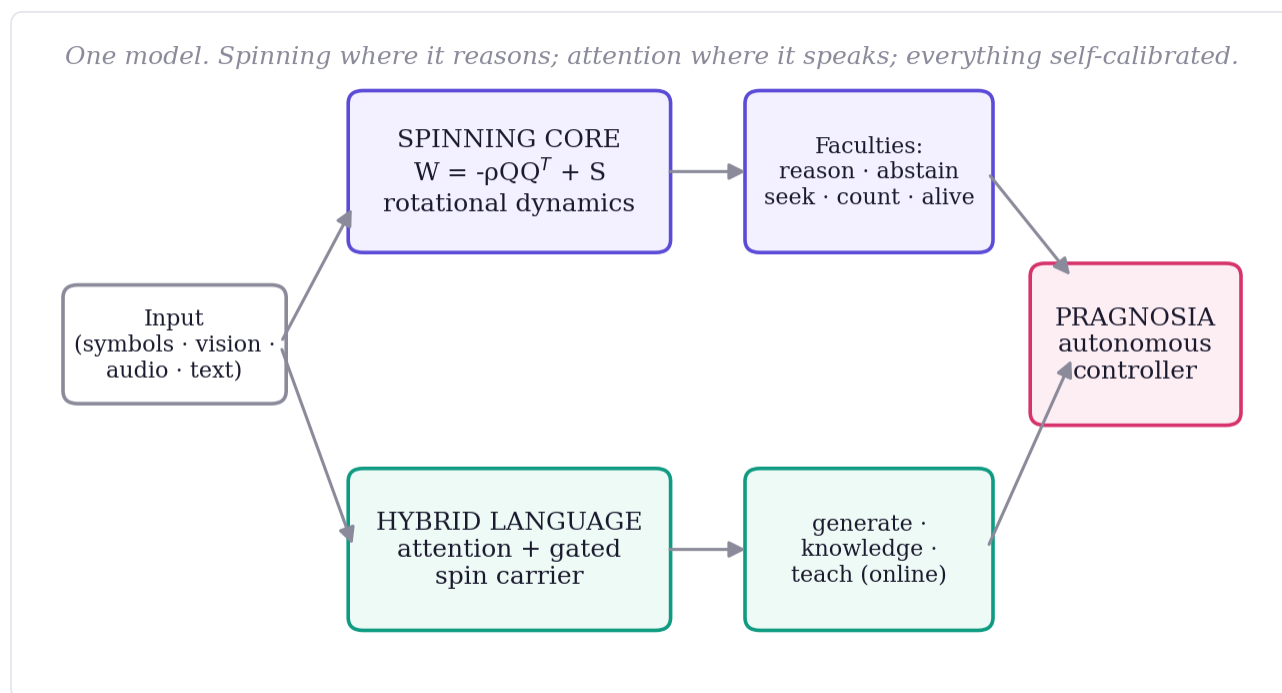


Figure 5. System overview. A shared input feeds the pure-spin reasoning core (top) and the attention+spin-carrier language faculty (bottom); both feed an autonomous controller that decides each action from the model's own signals.

Nothing hardcoded.

Every internal scale is derived from data and re-derived by a `recalibrate()` step as the model grows. Word importance is the self-information $-\log p(\text{token})$ estimated from the training corpus — function words emerge as low-weight, content words high — with no stopwords list. The abstention boundary is the high-percentile of the model's own uncertainty on familiar text. The memory-match threshold is read off the corpus's own background similarity distribution. None of these is hand-set.

Surprise-modulated plasticity.

Continual learning is self-paced like neuromodulation. When taught a fact, the brain sets its own learning rate from its own surprise — the prediction error divided by its familiarity boundary — learning hard on surprising input and gently on familiar input, and stopping once the fact ceases to surprise it (Fig. 5). Beyond biological plausibility, gentle steps on familiar input reduce interference. Taught facts persist across sessions; the model recalls them in fresh contexts (e.g., "the capital of Mongolia is" → "Ulaanbaatar") while retaining prior abilities.

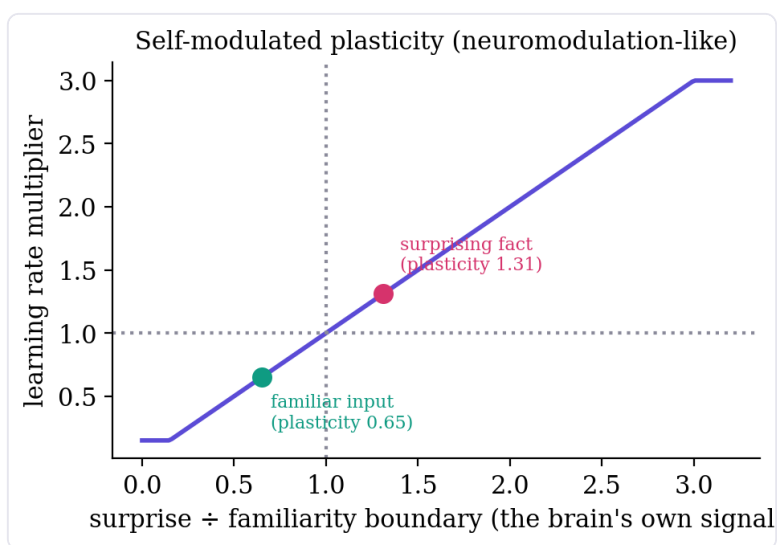


Figure 6. Self-modulated plasticity. The online learning rate is a function of the model's own surprise relative to its self-calibrated familiarity boundary; measured operating points for a surprising and a familiar input are marked.

Autonomous control.

A single controller decides, for any input, whether to answer, seek from memory, abstain, or learn — entirely from its own confidence and curiosity signals, with no keyword rules. Self-knowledge (the model's name and nature) is learned into the weights and recalled by the model's own generation, gated by its own confidence.

6. Experiments and Results

Table 1. The integrated self-test: one model, fourteen checks, all passing. (↓ lower is better.)

Capability	Measured	Criterion	Status
Reasoning ×3 skills (parity / sum / max)	0.92 / 0.90 / 0.93	>0.90	✓
P5 confidence gap (knowable vs. unanswerable)	0.34	>0.30	✓
P6 abstain: known↓ / unknowable	0.19 / 1.00	<0.25 / >0.60	✓
Language parse / generate (held-out)	1.00 / 1.00	>0.80	✓
Seek: query language / answer / random-ctrl↓	1.00 / 1.00 / 0.13	>0.90 / >0.85 / <0.45	✓
Exact accumulation, length 16	1.00	>0.90	✓
Alive loop: final / retention gap↓	1.00 / 0.00	>0.95 / <0.15	✓
Brain-swap (causal): swapped / input / random	0.59 / 0.10 / 0.21	swapped >> random	✓

Scaling the language faculty from 0.3M parameters and toy grammars to multi-hundred-million-parameter models on real corpora improves coherence and factual recall, while the full integrated self-test continues to pass — text training provably never touches the reasoning-faculty weights, which we verify after every change.

Trained instances.

The dense-tanh carrier was first verified at 176M parameters (d=1024, 12 layers, vocabulary 16,384), where validation perplexity fell monotonically with no divergence and the full integrated self-test held throughout. The parallel diagonal-complex carrier is now the production model: a 228M instance trains ~2× faster than that 176M baseline and reaches the same perplexity (20.76 vs. 20.47), with the brain-swap control still ordered own < swapped < random and the self-test at 14/14. At these scales the models are coherent at the sentence level, write simple correct code (def add(a,b): → return a + b), and acquire the *form* of step-by-step reasoning and encyclopedic text; single-step facts and arithmetic are real while multi-hop composition remains weak — the undertraining-for-size signature, expected before the compute-optimal token budget is reached. Continual learning works end-to-end at every scale: the model is taught a new fact, recalls it in a fresh context (e.g. "the capital of Mongolia is" → "Ulaanbaatar"), and retains prior abilities (no forgetting). The one honest weakness is language abstention: confidence-thresholded refusal catches clearly out-of-distribution inputs but is noisy on familiar-phrasing/unknown-content; a trained confidence head (the P6 recipe applied to language) is the intended sharpening.

7. Comparison with Large Language Models

Table 2. Where each paradigm wins. The two are built on opposite bets.

Property	Today's LLMs	Pragnosia
Fluent language / world knowledge	Strong / vast	Decent / limited (small)
Reasoning depth	Strong	Shallow but real
Causally-proven reasoning	Unverified	Yes (brain-swap)
Knows what it doesn't know	Partial, bolted on	Intrinsic (abstains)
Learn after training, no retrain	No (frozen)	Yes (persistent)
Forgetting on update	Catastrophic	Low (replay + plasticity)
Self-paced learning rate	Fixed schedule	Surprise-modulated
Hand-set behavior	—	None (self-derived)
Size / cost	Data-center	Single laptop GPU

An LLM is a brilliant but frozen library; Pragnosia is a tiny living mind. The bet is that the properties a frozen model lacks — being alive, honest about ignorance, and causally reasoning — are worth pursuing on a substrate designed for them, even at the cost of present-day breadth.

8. Limitations

We state these plainly, as every claim above carries a number. (i) The model is *undertrained for its grown size*: single-step reasoning, arithmetic, and fact recall are real, but multi-hop composition — relational analogy, multi-step deduction, systematic generalization — is weak. This is the undertraining signature; the lever is more training tokens, not more parameters or better prompting. (ii) Like all small models it can be *confidently wrong* on familiar-but-incorrect facts; its abstention catches the clearly-unknown, not the subtly-wrong. (iii) The *language* spin carrier is a diagonal-complex parallel scan (§4), $\sim 2\times$ faster at matched quality; the *pure-spin reasoning* core remains sequential by design (its non-associative tanh dynamics are what the brain-swap test verifies). (iv) Self-growth must be governed: an earlier plateau trigger over-grew the model to 1.4B (§4); the probe-confirmed-saturation rule fixes this but the resulting large checkpoint still needs the tokens to fill it. (v) Several non-language faculties remain at toy scale, wired and verified but not yet individually scaled. None of these is hidden; they are the honest frontier.

9. Future Directions and Applications

Roadmap.

The mechanisms are established; the remaining work is scale, data, and maturation of each faculty. *Near-term*: replace the calibrated-threshold abstention with a trained confidence head (the proven two-phase recipe applied to language) for sharper refusal, and — the dominant lever — enlarge the corpus with reasoning-rich data, since reasoning is learned from examples rather than engineered. *Mid-term*: scale the living faculties — continual learning from a handful of facts to thousands with rehearsal-based consolidation, language-mediated seeking against a real retrieval index (RAG-style), and relational vision and audio beyond toy grids and tones. *Long-term*: a single on-device model that runs continuously, learns its user over months without forgetting, abstains rather than fabricates, and can explain its reasoning — an always-on, lifelong, personal mind rather than a frozen oracle.

Applications.

The compelling use cases are precisely those a frozen, cloud-bound model cannot serve. (i) *On-device assistants that learn the user*: a small, private model that continually acquires a person's facts and preferences locally, with

no retraining and no data leaving the device. (ii) *Honest assistance in high-stakes domains* (medicine, law, finance): intrinsic abstention — saying "I do not know" and seeking — is safer than fluent fabrication. (iii) *Lifelong-learning robotics and embedded agents*: curiosity-driven online learning from the environment without catastrophic forgetting, within a footprint that fits on the edge. (iv) *Auditable AI*: causally-probeable reasoning (brain-swap) for accountability-sensitive settings. (v) *Adaptive tutoring and personalization*: a model that adapts to an individual learner over time rather than serving a static average. In each case the differentiator is not breadth of knowledge but the trio of properties this substrate is designed for — alive, honest, and causal — at a cost that runs anywhere.

10. Conclusion

Pragnosia demonstrates that a single small model can reason in a causally-verifiable way, model its own uncertainty and abstain, seek what it lacks, learn continually without forgetting under self-modulated plasticity, and decide its own actions — with no hand-set constants — while a spin-attention hybrid recovers usable language. These are precisely the brain-like properties that frozen large models lack. The path forward is not architectural cleverness but scale and reasoning-rich data on the same substrate; the mechanisms are in place and the integration is complete. We release the full system, every measured number, and every limitation.

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